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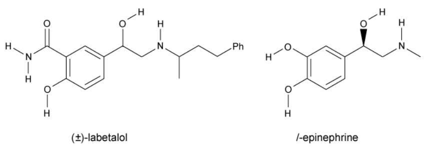

Figure 1 Structure of (±)-labetalol and *l*-epinephrine

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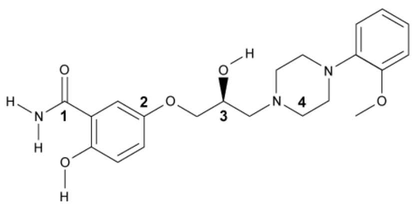
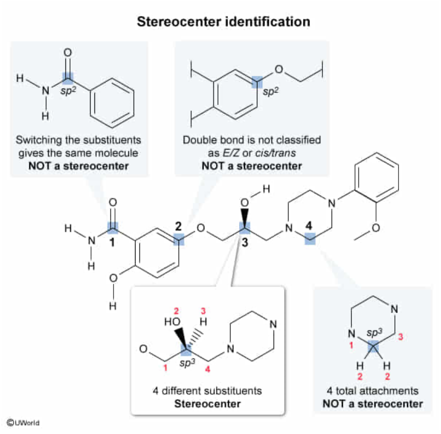


What position(s) on the analog of labetalol shown above can be classified as a stereocenter?

- A. Position 1 only [3%]
- B. Positions 2 and 3 only [4%]
- ✔ C. Position 3 only [87%]
- D. Positions 3 and 4 only [4%]

Correct
88% answered correctly

Explanation:



A stereocenter is an atom where when two substituents switch positions, a new stereoisomer is formed. Both sp^3 and sp^2 hybridized carbons can be stereocenters. An ***sp*³** hybridized carbon that is attached to **four different groups** is called a chiral center. If the central carbon is bonded to two or more identical atoms, subsequent atoms should be compared. This process should be repeated until a difference is found or the chains end. If no difference is found, the four groups to which the carbon is attached are not all different and the central carbon is not a stereocenter.

The ***sp*²** hybridized carbons in an alkene are stereocenters if the **alkene** can be classified as *E*/*Z* or *cis*/*trans*. Terminal alkenes (ie, a C=C bond at the end of a carbon chain) are *not* stereocenters because the sp^2 carbon at the end of the chain is bonded to two H atoms; therefore, the terminal alkene is not *E*/*Z* or *cis*/*trans*.

The carbon at position 3 is sp^3 hybridized and bonded to two carbon atoms, an oxygen atom, and a hydrogen atom. One of the carbon substituents is bonded to oxygen and the other is bonded to nitrogen, so the two carbon

substituents are different and the central carbon is bonded to four unique groups. Therefore, the carbon at position 3 is a stereocenter.

(Choice A) Although position 1 is an sp^2 hybridized carbon, it is a carbonyl (C=O) and *cannot* be classified as a stereocenter. Switching the two substituents on the carbon atom does not change the molecule (ie, it is the same molecule and *not* a stereoisomer).

(Choice B) Position 2 is an sp^2 hybridized carbon that is part of an aromatic ring. Carbon atoms in aromatic rings are not classified as *E*/*Z* or *cis*/*trans*, and therefore are *not* stereocenters.

(Choice D) Position 4 contains only three unique attachments because two of the four attach to hydrogen atoms. Therefore, it is *not* a stereocenter.

Educational objective:

A stereocenter is any atom where when two substituents switch positions, a stereoisomer is formed. An sp^3 carbon is a stereocenter if it is bonded to four unique groups. The sp^2 carbons in alkenes are stereocenters if the alkene can be classified as *E*/*Z* or *cis*/*trans*.

Time spent: 0 sec
Subject: Organic Chemistry

QID: 401034
System: Introduction to Organic Chemistry

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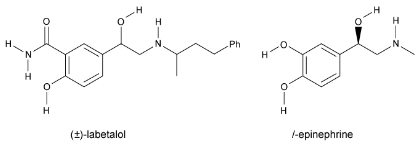

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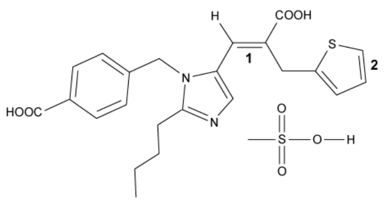
The structure of eprosartan mesylate is shown below. Double bonds 1 and 2, respectively, can be classified as:

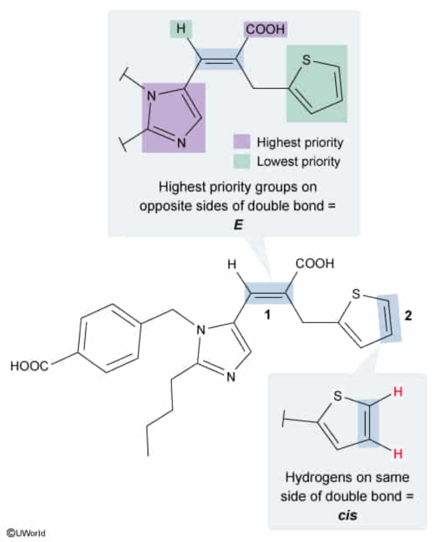


- A. *E* for double bond 1 and *cis* for double bond 2 [65%]
- B. *E* for double bond 1 and *trans* for double bond 2 [3%]
- C. *Z* for double bond 1 and *cis* for double bond 2 [29%]
- D. *Z* for double bond 1 and *trans* for double bond 2 [1%]

Correct

65% answered correctly

Explanation:



Double bonds, which are di–, tri–, or tetrasubstituted, can be classified two ways:

- *E*/*Z*
- *cis*/*trans*

The Cahn–Ingold–Prelog priority rules are used to assign priority to the substituents on each carbon to determine whether a double bond is *E* or *Z*. Whenever the higher priority groups of each carbon are on **opposite sides** of the double bond, it is considered ***E***, and it is ***Z*** when the **higher priority** groups are on the **same side**. Disubstituted double bonds are a special case in which an *E* bond is called *trans* and a *Z* bond is called *cis*.

Double bond 1 contains a trisubstituted double bond. On the left side of the double bond, the ring is higher priority

because carbon has a greater atomic number than hydrogen. The right side of the double bond is attached to two carbon atoms:

- a carboxyl carbon (COOH), which has three bonds to oxygen (ie, the double bond to oxygen counts as two bonds)
- a methylene group (–CH_2), which has two bonds to hydrogen and one bond to a carbon in a ring

Comparing the atom with the highest molecular weight bonded to the carboxyl and methylene carbons (ie, oxygen and carbon), oxygen has a greater atomic weight than carbon, and thus the carboxyl group has a higher priority. The high-priority groups are on opposite sides of the double bond, so it is classified as *E*.

Double bond 2 is a disubstituted double bond in which both hydrogen atoms (or both high-priority groups) are on the same side; therefore, it is *cis*.

(Choices B, C, and D) For double bond 2 to be *trans*, the hydrogen atoms (and the high-priority groups) would have to be on opposite sides of the double bond. For double bond 1 to be *Z*, the high-priority groups would have to be on the same side of the double bond.

Educational objective:
E and *Z* designations distinguish stereoisomers about double bonds, in which *Z* signifies high-priority groups on the same side of the bond and *E* indicates the high-priority groups on opposite sides of the bond. In disubstituted double bonds, the *Z* conformation is also called *cis* and the *E* conformation is called *trans*.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401035
System:
Introduction to Organic Chemistry

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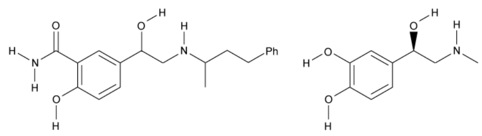

(±)-labetalol   *l*-epinephrine

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(*S,R*)- and (*R,R*)-labetalol, the active forms of the drug, can be described as which of the following?

I. Diastereomers

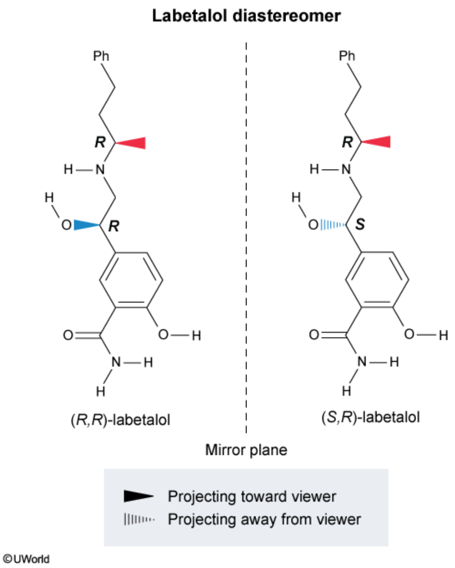
II. Enantiomers

III. Conformational isomers

- A. I only [80%]
- B. I and II only [4%]
- C. II and III only [11%]
- D. I, II, and III [3%]

Correct
79% answered correctly

Explanation:



Stereoisomers are molecules that contain the same atom connectivity but differ in the spatial arrangements of the atoms. **Diastereomers** are stereoisomers that are **not mirror images**. These molecules contain *at least* two stereocenters in which **one or more** (but **not all**) of the **stereocenters** on corresponding carbon atoms are in **opposite configurations**.

(*S*,*R*)- and (*R*,*R*)-labetalol are stereoisomers that are not mirror images, and each compound contains two stereocenters. The two compounds have different configurations at the stereocenter in position 1 (one *S* and one *R*) and have the same configuration at the stereocenter in position 2 (both *R*). Therefore, (*S*,*R*)- and (*R*,*R*)-labetalol are diastereomers **(Number I)**.

(Number II) Enantiomers are stereoisomers that are nonsuperimposable mirror images. These molecules contain one or more stereocenters, and each stereocenter is the opposite configuration from the corresponding stereocenter in the other molecule. For example, the enantiomer of (*S*,*R*)-labetalol is (*R*,*S*)-labetalol.

(Number III) Conformational isomers are different forms of the same molecule that are generated as atoms rotate about their bonds. Unlike other stereoisomers, conformational isomers can rapidly interconvert by rotation without the need to break any bonds.

Educational objective:
Diastereomers are stereoisomers that are not mirror images. They contain at least two stereocenters in which one or more (but not all) are in opposite configurations.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401036
System:
Introduction to Organic Chemistry

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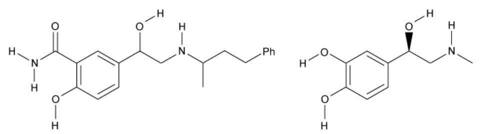

(±)-labetalol   *l*-epinephrine

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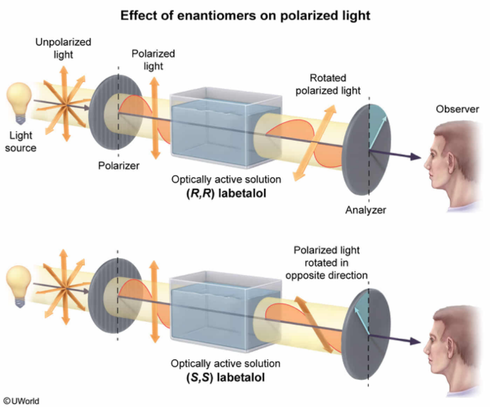
Would (*R,R*)- and (*S,S*)-labetalol be expected to rotate plane-polarized light in the same direction?

- A. Yes, because they are diastereomers. [4%]
- B. Yes, because the rotations would have the same magnitude. [3%]
- C. No, because they are enantiomers. [70%]
- D. No, because the rotations would cancel out each other. [21%]

Correct
69% answered correctly

Explanation:



Normal light is made up of waves that vibrate in all planes perpendicular to the direction of travel. When the light meets a polarizer, only waves that are aligned with the polarizer can pass through, so all the light that passes through vibrates in one direction. This resulting light is known as plane-polarized (linear) light.

When chiral molecules interact with polarized light, the interactions cause the angle of oscillation to rotate. An *R* or *S* configuration is not predictive of the direction of rotation. Every molecule is different, and the concentration and length of the path through which the light travels affect the angle of rotation; the direction and magnitude of rotation can only be determined experimentally. Considering experimental parameters such as concentration and path length, the observed rotation can be standardized with the calculation of specific rotation. **Enantiomers** always have **specific rotations** of **equal magnitude** but **opposite directions**. (*R*,*R*)-labetalol and (*S*,*S*)-labetalol are enantiomers, and therefore rotate plane-polarized light by the same amount but in opposite directions.

(Choice A) (*R,R*)- and (*S,S*)-labetalol are enantiomers, not diastereomers. *No correlation* exists between *diastereomers* and the magnitude or direction of plane-polarized light rotation.

(Choice B) If two samples rotated plane-polarized light by the same magnitude and direction, they would likely be the same compound. Although enantiomers (*R,R*)- and (*S,S*)-labetalol rotate plane-polarized light by the same magnitude, they cause the light to rotate in opposite directions.

(Choice D) If a sample were to have a *rotation of zero*, it would mean that the rotations have canceled each other out. This occurs when the sample is a 50:50 mixture of enantiomers (ie, 50% *R* and 50% *S*) known as a *racemic mixture*.

Educational objective:
Enantiomers rotate plane-polarized light by the same magnitude but in opposite directions. No correlation exists between diastereomers and the magnitude or direction of rotation. Racemic mixtures have a

rotation of zero because they consist of equal amounts of each enantiomer.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401037
System:
Introduction to Organic Chemistry

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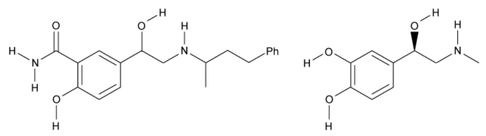

(±)-labetalol   *l*-epinephrine

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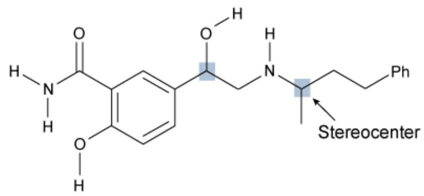
The number of stereoisomers possible for a given molecule is 2^n. The variable *n* denotes the number of:

- A. enantiomers.
- ✔ B. stereocenters. [97%]
- C. diastereomers.
- D. epimers.

Correct
96% answered correctly

Explanation:

2^n = Number stereoisomers
n = Number stereocenters



Labetalol
2^2 = 4 Stereoisomers

Stereoisomers are molecules that differ in spatial arrangement but have the same connectivity and molecular formula. The number of stereoisomers for a particular compound corresponds to the number of configurations possible for that compound and is directly related to the number of stereocenters in the molecule. Each stereocenter can adopt two different conformations, so the expression $\mathbf{2^n}$ provides the **maximum number of stereoisomers** possible for the particular compound, where ***n*** is the **number of stereocenters** in the molecule. For example, labetalol has two stereocenters, and therefore 2^2 = 4 possible stereoisomers. A compound with three stereocenters has 2^3 = 8 possible configurations, and so on.

(Choices A and C) Enantiomers and diastereomers are types of stereoisomers, and are included in the number of total stereoisomers that can exist.

(Choice D) Epimers are a type of diastereomer that differs in configuration at only one stereocenter. The number of epimers is not factored into the calculation of the number of stereoisomers possible for a particular molecule.

Educational objective:

A molecule with n stereocenters can have a maximum of 2^n stereoisomers.

Time spent:0 sec	**QID**:401038
Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry

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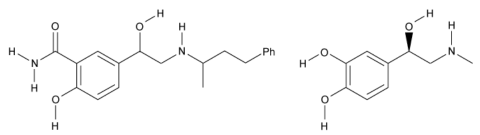
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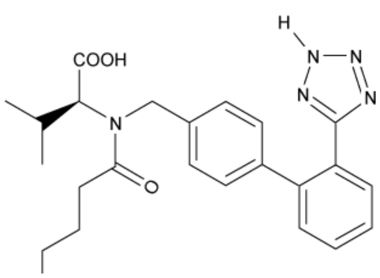
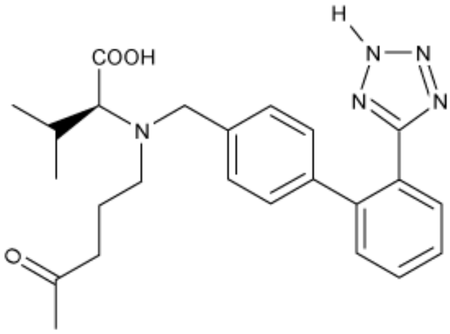
Below is the structure of valsartan. All the following are structural isomers of valsartan EXCEPT:



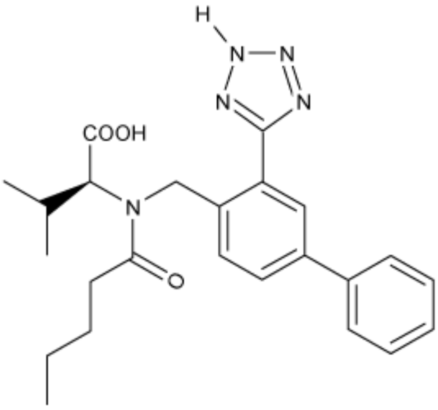
|

○ A.



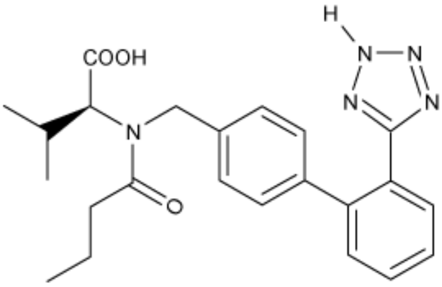
[6%]

○ B.



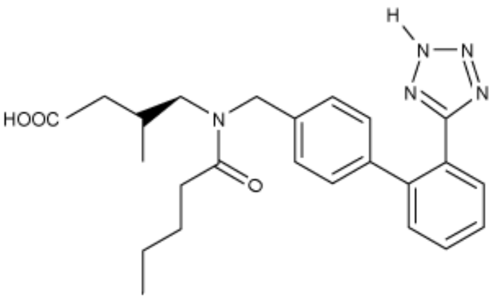
[5%]

✔ ◉ C.



[78%]

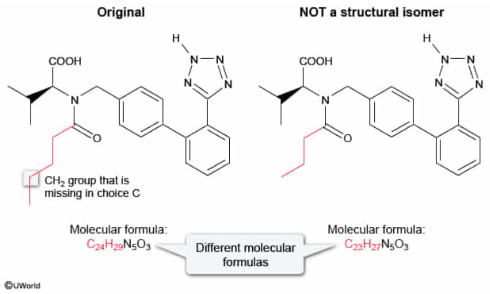
○ D.



[10%]

Correct
77% answered correctly

Explanation:



Structural isomers, also known as **constitutional isomers**, have the **same molecular formula** and thus the same molecular weight but **differ** in the order of **atom connectivity** in at least one position. Because a methylene group (–CH_2–) is missing from the carbonyl-containing (C=O) alkyl chain in Choice C, this structure does not have the same molecular formula as the original compound, and therefore is not a structural isomer of valsartan.

The other available choices possess the same molecular formula as valsartan. The only differences are that each compound has one or more functional groups in a different location than in the original molecule, making each a structural isomer of valsartan.

- A carbonyl group was moved in **Choice A**.
- An aromatic hydrogen atom and a tetrazole (nitrogen-containing ring) were moved in **Choice B**.
- A carboxyl group and a hydrogen atom were moved in **Choice D**.

Educational objective:
A structural isomer, or constitutional isomer, is a molecule that contains the same molecular formula as another molecule. The atoms in each molecule differ in their connectivity.

Time spent:0 sec
Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry